

Program 7 :

Step 1: Start Minikube

Step 2: Check Minikube Status

```
rv24mc028_chethanacs@Elite:~/DEVops/prg7$ minikube start
🌟 minikube v1.37.0 on Ubuntu 22.04
👉 Using the docker driver based on existing profile
👉 Starting "minikube" primary control-plane node in "minikube" cluster
🚚 Pulling base image v0.0.48 ...
🔧 Updating the running docker "minikube" container ...
❗ Failing to connect to https://registry.k8s.io/ from inside the minikube container
💡 To pull new external images, you may need to configure a proxy: https://minikube.sigs.k8s.io/docs/reference/networking/proxy/
🐳 Preparing Kubernetes v1.34.0 on Docker 28.4.0 ...
🔍 Verifying Kubernetes components...
   ■ Using image gcr.io/k8s-minikube/storage-provisioner:v5
🌞 Enabled addons: storage-provisioner, default-storageclass
🏠 Done! kubectl is now configured to use "minikube" cluster and "default" namespace by default
rv24mc028_chethanacs@Elite:~/DEVops/prg7$ minikube status
minikube
type: Control Plane
host: Running
kubelet: Running
apiserver: Running
kubeconfig: Configured
```

Step 3: Create Deployment File

```
rv24mc028_chethanacs@Elite:~/DEVops/prg7$ cat nginx-deployment.yaml
apiVersion: apps/v1
kind: Deployment
metadata:
  name: nginx-deployment
spec:
  replicas: 2
  selector:
    matchLabels:
      app: nginx
  template:
    metadata:
      labels:
        app: nginx
    spec:
      containers:
      - name: nginx
        image: nginx:latest
        ports:
        - containerPort: 80
rv24mc028_chethanacs@Elite:~/DEVops/prg7$
```

Step 4: Apply the Deployment

```
rv24mc028_chethanacs@Elite:~/DEVops/prg7$ kubectl apply -f nginx-deployment.yaml
deployment.apps/nginx-deployment configured
rv24mc028_chethanacs@Elite:~/DEVops/prg7$
```

Step 5: Check Deployment and Pod Status

```
rv24mc028_chethanacs@Elite:~/DEVops/prg7$ kubectl get deployments
NAME                READY    UP-TO-DATE    AVAILABLE    AGE
nginx-deployment    2/2      2              2             25d
rv24mc028_chethanacs@Elite:~/DEVops/prg7$ kubectl get pods
NAME                                READY    STATUS      RESTARTS      AGE
backup-job-29421230-qq58w           0/1      Completed   0              16d
backup-job-29445700-vfpnn           0/1      Completed   0              5m20s
backup-job-29445705-27jt2           0/1      Completed   0              2m36s
kubia                                1/1      Running     3 (3m33s ago)  17d
my-cron-29445705-x4k6z              0/1      Completed   0              2m36s
my-cron-29445706-xrd78              0/1      Completed   0              96s
my-cron-29445707-pbjw2              0/1      Completed   0              36s
nginx-deployment-6475cd8bfc-cjkck    1/1      Running     4 (3m43s ago)  25d
nginx-deployment-6475cd8bfc-qqnxs    1/1      Running     4 (3m43s ago)  25d
simple-job-zjn46                     0/1      Completed   0              25d
```

Step 6: Expose the Deployment as a Service

Step 7: Check Services

```
rv24mc028_chethanacs@Elite:~/DEVops/prg7$ kubectl expose deployment nginx-deployment --type=NodePort --port=80
Error from server (AlreadyExists): services "nginx-deployment" already exists
rv24mc028_chethanacs@Elite:~/DEVops/prg7$ kubectl get services
NAME                TYPE                CLUSTER-IP      EXTERNAL-IP      PORT(S)          AGE
kubernetes          ClusterIP           10.96.0.1        <none>            443/TCP           30d
kubia                NodePort            10.97.179.14     <none>            8080:30252/TCP    7d
kubia-http          LoadBalancer       10.104.6.56      <pending>         8080:30251/TCP    8d
mypod               ClusterIP           10.96.179.182    <none>            80/TCP            5d
nginx-deployment    NodePort            10.103.155.14    <none>            80:31729/TCP      5d
web                 ClusterIP           10.99.243.59     <none>            80/TCP            5d
rv24mc028_chethanacs@Elite:~/DEVops/prg7$ minikube service nginx-deployment
```

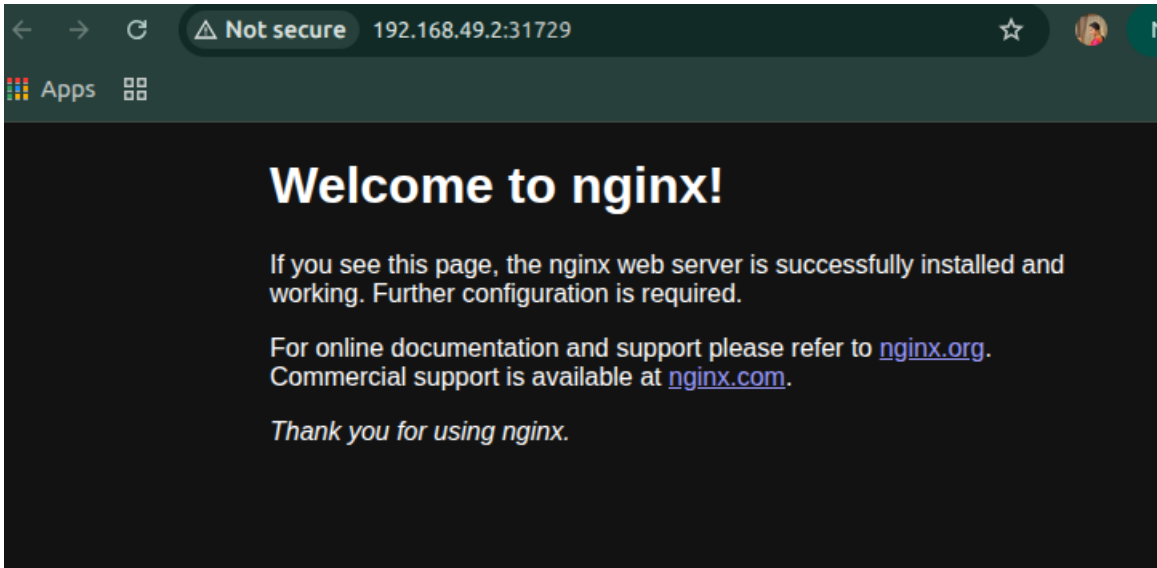
NAMESPACE	NAME	TARGET PORT	URL
default	nginx-deployment	80	http://192.168.49.2:31729

 Opening service default/nginx-deployment in default browser...

Step 8 : Access the Web Application

```
rv24mc028_chethanacs@Elite:~/DEVops/prg7$ Opening in existing browser session.
^C
rv24mc028_chethanacs@Elite:~/DEVops/prg7$ kubectl delete pods --all
pod "backup-job-29445700-vfpnn" deleted from default namespace
```

Output :



Step 9:
Delete
all
pods ,

services and deployments

```
rv24mc028_chethanacs@Elite:~/DEVops/prg7$ kubectl delete pods --all
pod "backup-job-29445700-vfpnn" deleted from default namespace
pod "backup-job-29445705-27jt2" deleted from default namespace
pod "backup-job-29445710-stv7l" deleted from default namespace
pod "kubia" deleted from default namespace
pod "my-cron-29445709-vqs66" deleted from default namespace
pod "my-cron-29445710-ng75j" deleted from default namespace
pod "my-cron-29445711-r7md9" deleted from default namespace
pod "nginx-deployment-6475cd8bfc-cjkck" deleted from default namespace
pod "nginx-deployment-6475cd8bfc-qqn8" deleted from default namespace
pod "simple-job-zjn46" deleted from default namespace
rv24mc028_chethanacs@Elite:~/DEVops/prg7$ kubectl delete svc --all
service "kubernetes" deleted from default namespace
service "kubia" deleted from default namespace
service "kubia-http" deleted from default namespace
service "mypod" deleted from default namespace
service "nginx-deployment" deleted from default namespace
service "web" deleted from default namespace
rv24mc028_chethanacs@Elite:~/DEVops/prg7$ kubectl delete deployments --all
deployment.apps "nginx-deployment" deleted from default namespace
```